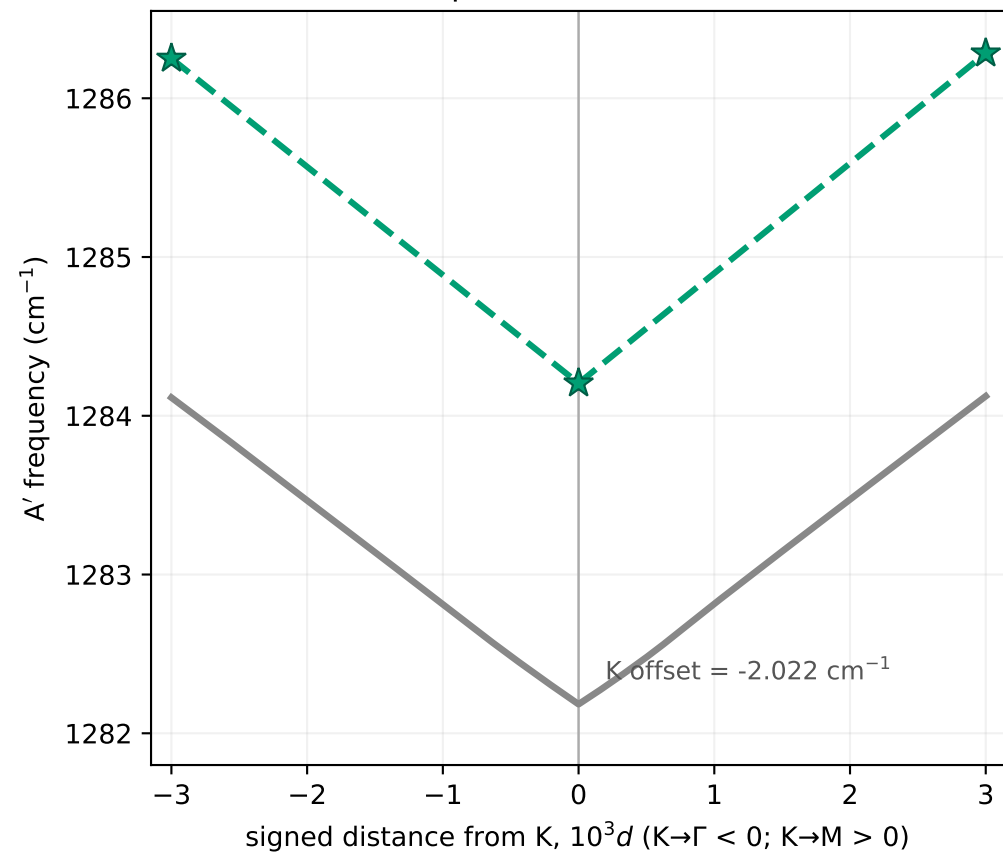
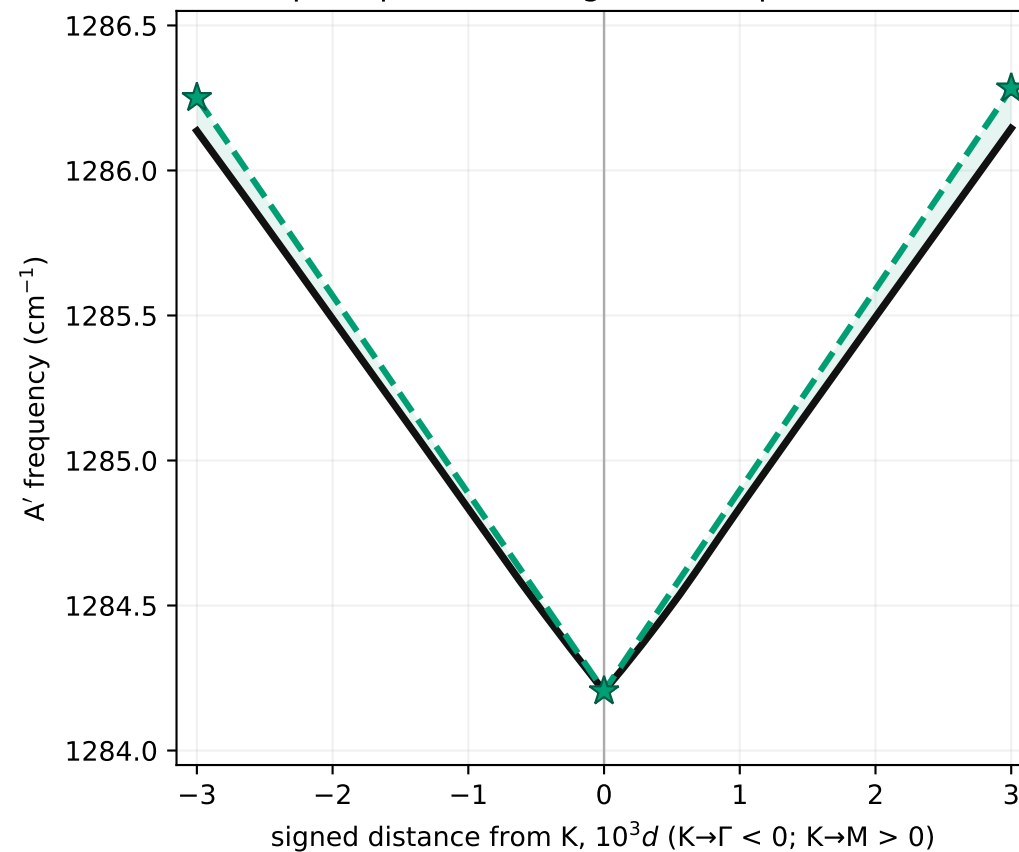


Graphene zero-smearing K cusp: extrapolated DFPT curve vs MLIP + long-range correction

(a) Absolute frequencies with current calibration



(b) Cusp shape after using the extrapolated K anchor



smearing/degauss = 0 Ry

- MLIP + full EPC LR (original k288 K anchor)
- same MLIP + LR curve reanchored to DFPT K_∞
- - - local DFPT $N \rightarrow \infty$ cusp curve
- ★ fixed-q DFPT $N \rightarrow \infty$ anchors

DFPT K_∞ : 1284.205 cm^{-1}
 Original model K: 1282.183 cm^{-1}
 Common reanchoring shift: $+2.022 \text{ cm}^{-1}$

$d=0.003$ cusp depth (KG / KM):
 DFPT $N \rightarrow \infty$: $2.046 / 2.079 \text{ cm}^{-1}$
 MLIP + LR: $1.932 / 1.940 \text{ cm}^{-1}$
 Signed error: $-5.57\% / -6.65\%$

DFPT dashed curve is supported only on $|d| \leq 0.003$.
 It is piecewise linear through three fixed-q intercepts.